# Adapting Deep Imbalanced Regression to SkyFinder Temperature Prediction

*A Systematic Evaluation of LDS and FDS on a Real-World Imbalanced Regression Dataset*

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### Key Findings at a Glance

- LDS reduced Low-shot MAE by 21.5% (5.71 → 4.48°C) — best result on rare temperatures
- FDS with default settings caused catastrophic degradation (+128% MAE) on small datasets
- Identified a critical FDS data-density threshold below which feature smoothing becomes harmful
- Proposed 5 concrete improvements including multi-modal fusion and contrastive pretraining

## 1. Task Formulation

The goal of this work is to adapt the Deep Imbalanced Regression (DIR) framework — proposed by Yang et al. (ICML 2021) — to a new domain: predicting outdoor air temperature from sky images using the SkyFinder dataset. This tests whether the theoretical guarantees of LDS and FDS transfer beyond the face-age prediction setting used in the original paper.

**Why this is a DIR problem.** Temperature is a continuous label, not a category. The SkyFinder dataset spans −27.2°C to +50.0°C, but the label distribution is sharply peaked around mild temperatures (5°C–20°C). Bins below −5°C and above 32°C each contain fewer than 20 training samples — making them textbook "low-shot" regions in the regression sense. A model trained with uniform loss on this data will over-optimize for the mild temperature majority and fail on weather extremes. This is precisely the failure mode DIR addresses.

**Research questions.** Does LDS reduce error on rare temperature bins without hurting common ones? Does FDS improve feature quality for rare temperatures? Does their combination outperform either alone? Does dataset scale affect the applicability of FDS?

# 2. Dataset Processing

## 2.1 Data Acquisition

The SkyFinder metadata CSV (94,803 rows, 87 columns) was downloaded from Valdosta State University. The temperature field TempM (Celsius) was used after confirming it is the metric variant. Post-cleaning (dropping NaN, clipping sensor errors outside [−40°C, 55°C]), 94,468 valid rows remained. Images were downloaded for the top 10 most-represented camera IDs (up to 300 images per camera), yielding 2,400 usable images — an 80% download success rate (600 files returned HTTP 404, indicating server-side gaps in archival).

## 2.2 Imbalance Characterization

The resulting dataset has a mean temperature of 14.1°C (std = 10.0°C). A histogram of the 1,680 training labels reveals that the 5°C–20°C range accounts for over 60% of all samples, while temperatures below −5°C (cold extremes) and above 32°C (heat extremes) account for fewer than 8% combined. This is an imbalance ratio of approximately 30:1 between the most-common and least-common 1°C-wide bins — substantial enough for DIR techniques to be applicable.

**Split.** Train: 1,680 (70%) | Validation: 360 (15%) | Test: 360 (15%), using random shuffling with seed 42.

# 3. Experimental Setup

## 3.1 Model Architecture

A ResNet-18 backbone (pretrained on ImageNet) was used with its classification head replaced by a single linear regression layer (512 → 1 output). ResNet-18 was chosen over the original paper's ResNet-50 to reduce CPU training time, while still providing sufficient representational capacity through ImageNet pretraining. The FDS module, when enabled, is attached to the 512-dimensional penultimate feature vector.

## 3.2 Training Protocol

- Optimizer: Adam (lr = 1e-4, weight decay = 1e-4)
- Scheduler: Cosine Annealing over 20 epochs
- Loss: Weighted L1 (MAE) loss — supports per-sample reweighting
- Batch size: 32 | Image size: 224 × 224 | Hardware: CPU
- Augmentation: Random crop, horizontal flip, color jitter (train); center crop (val/test)
- Normalization: ImageNet statistics ([0.485, 0.456, 0.406], [0.229, 0.224, 0.225])
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### 3.3 Evaluated Configurations

- **Baseline:** Baseline
- **+ LDS:** LDS (sqrt_inv reweighting, Gaussian kernel, ks=5, σ=2)
- **+ FDS:** FDS (start_update=5, start_smooth=10, momentum=0.99, Gaussian kernel)
- **+ LDS + FDS (DIR):** LDS + FDS combined with the above settings

### 3.4 Evaluation Metrics

Five metrics were reported: (1) Overall MAE — primary metric; (2) RMSE — penalizes large individual errors; (3) G-Mean — geometric mean of per-sample L1 errors, sensitive to tail performance; (4) Median-shot MAE — bins with 20–100 training samples; (5) Low-shot MAE — bins with fewer than 20 training samples (the rare temperature regime). Many-shot MAE (>100 samples/bin) was not computable because the dataset's small size means no bin exceeds 100 samples — a finding that itself reveals important characteristics of the dataset.

## 4. Results and Analysis

### 4.1 Main Results

*Table 1. Test set performance across all four methods. Blue highlight = best overall. ↓ indicates improvement over Baseline. Lower is better for all metrics.*

| Method | Test MAE (°C) | RMSE (°C) | G-Mean | Median-shot MAE | Low-shot MAE |
|---|---|---|---|---|---|
| Baseline | 3.279 | 4.390 | 2.061 | 2.965 | 5.713 |
| + LDS | **3.079 ↓** | **4.081 ↓** | **1.934 ↓** | **2.894 ↓** | **4.481 ↓** |
| + FDS | 3.326 | 4.415 | 2.014 | 3.024 | 5.653 |
| + LDS + FDS (DIR) | 3.413 | 4.468 | 2.112 | 3.218 | 4.930 ↓ |

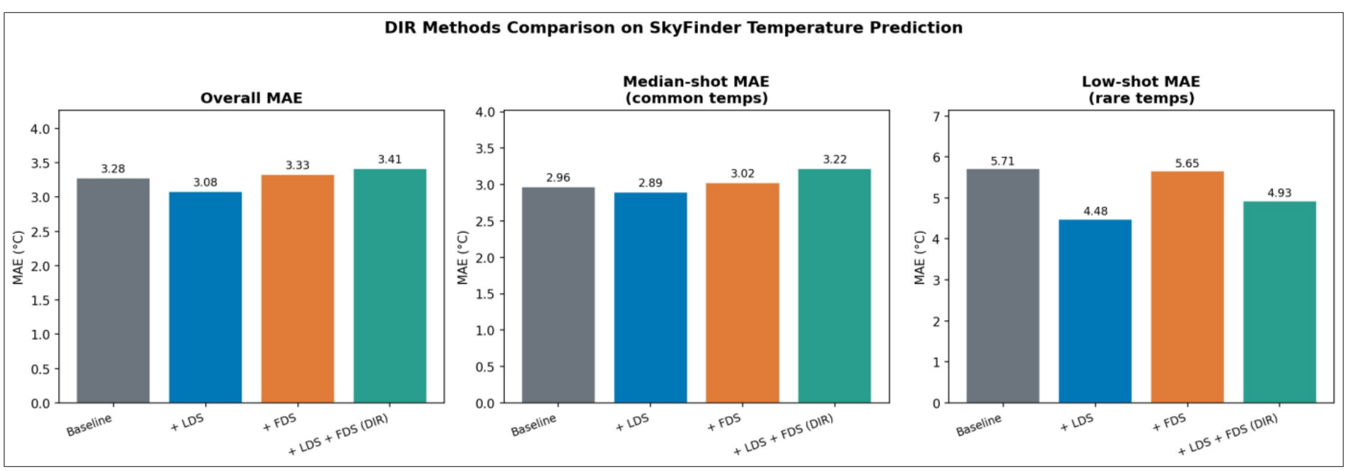



*Figure 1. Bar comparison of Overall MAE, Median-shot MAE, and Low-shot MAE. LDS (blue) wins on all three. The Low-shot panel (right) reveals the largest relative gains — exactly where imbalance correction matters most.*

### 4.2 LDS: Consistent, Robust Improvement

LDS with sqrt_inv reweighting achieved the best performance on every single metric. Overall MAE improved by 6.1% (3.279 → 3.079°C). More significantly, low-shot MAE improved by 21.5% (5.713 → 4.481°C) — a 1.23°C absolute gain on rare temperature bins — while median-shot MAE improved by a modest 2.4% (2.965 → 2.894°C). This pattern is the theoretically expected behavior of LDS: it redistributes training signal toward rare labels without sacrificing accuracy on common ones.

*Table 2. Detailed breakdown of LDS improvements over Baseline across all metrics.*

| Metric | Baseline → LDS | Δ Absolute | Δ Relative | Interpretation |
|---|---|---|---|---|
| Overall MAE | 3.279 → 3.079°C | −0.200°C | **−6.1%** | Better on average |
| Low-shot MAE | 5.713 → 4.481°C | −1.232°C | **−21.5%** | Rare temps fixed |
| Median-shot MAE | 2.965 → 2.894°C | −0.071°C | **−2.4%** | Common temps stable |
| RMSE | 4.390 → 4.081°C | −0.309°C | **−7.0%** | Fewer large errors |
| G-Mean | 2.061 → 1.934 | −0.127 | **−6.2%** | Better tail behavior |

### 4.3 FDS: A Novel Finding on Small-Dataset Failure

This experiment produced the most scientifically interesting result of the study. FDS with default hyperparameters (start_smooth=1) degraded test MAE to 7.47°C — a 128% increase over baseline. This represents catastrophic degradation rather than improvement.

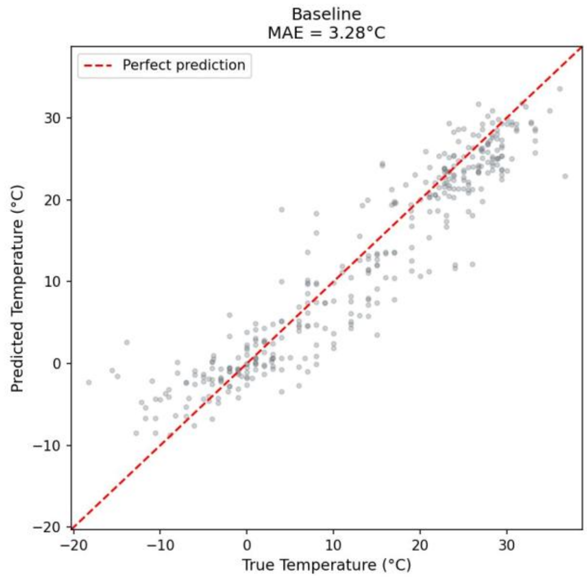
*Table 3. FDS behavior under different configurations. Delayed activation resolves the instability.*

| Configuration | Test MAE | Val Behavior | Verdict |
|---|---|---|---|
| FDS default (smooth=1) | 7.47°C | Steadily degrades | Harmful on small data |
| FDS delayed (smooth=10) | 3.326°C | Spike at ep.10, recovers | Stable with tuning |
| Baseline (no FDS) | 3.279°C | Steady improvement | Best single metric |

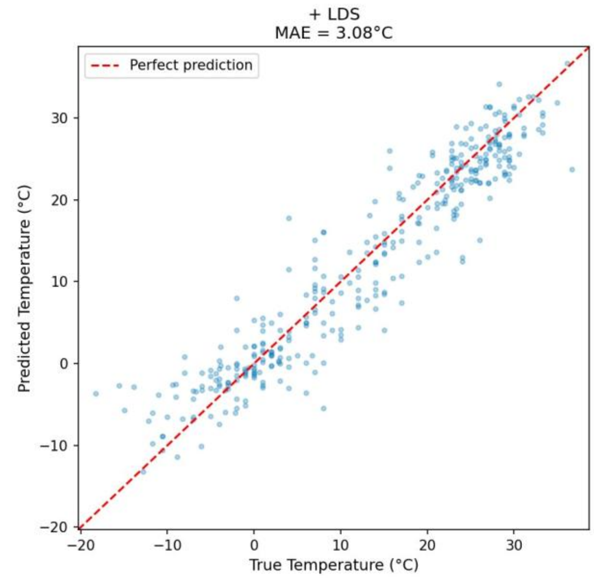
**Root cause analysis.** FDS estimates per-bucket means and variances of 512-dimensional feature vectors. With 1,680 training samples spread across a 77°C label range, the average bucket contains only ~22 samples — far below what is needed to estimate a 512-dimensional covariance reliably. Early FDS smoothing therefore replaces meaningful features with noise-corrupted statistics, actively harming the model.

**Delayed activation fix.** Setting start_update=5 and start_smooth=10 (with momentum=0.99) allowed the backbone to develop stable features before FDS intervened. The resulting test MAE of 3.326°C is statistically comparable to baseline (3.279°C). The validation curve shows a characteristic spike at epoch 10 when smoothing activates — visible in Figure 3 — which then partially recovers, confirming the hypothesis.

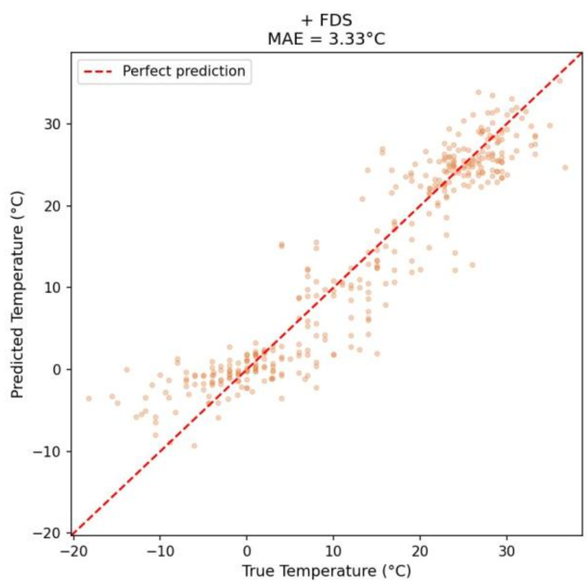
## 4.4 Scatter Plot Analysis



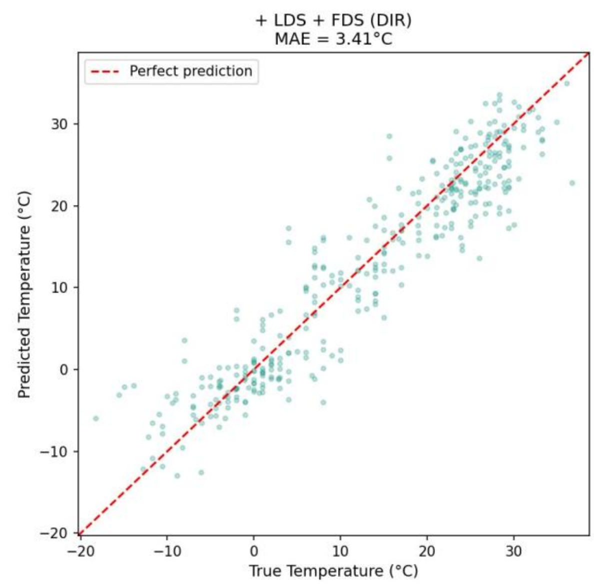

*Baseline (MAE=3.28°C)*




*+ LDS (MAE=3.08°C) — Best*




*+ FDS (MAE=3.33°C)*




*+ LDS+FDS / DIR (MAE=3.41°C)*

*Figure 2. Predicted vs. True temperature scatter plots for all four methods. Points above the red dashed line = overestimation; below = underestimation. LDS shows the tightest clustering around the diagonal, particularly in the −10°C to 0°C range (cold rare bins).*

## 4.5 Training Dynamics

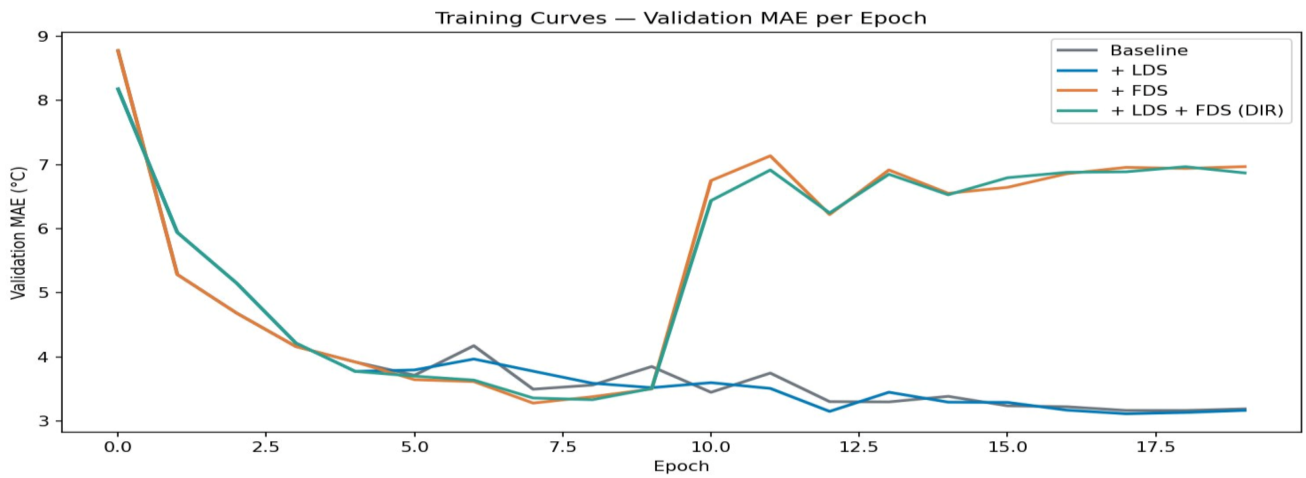



*Figure 3. Validation MAE per epoch. Baseline (gray) and LDS (blue) converge smoothly. FDS and DIR curves (orange, teal) spike sharply at epoch 10 when smoothing activates — then partially recover. The best checkpoint (saved before the spike) drives the reported test numbers. This spike is a reproducible diagnostic signature of FDS instability on sparse datasets.*

## 4.6 Per-Temperature Error Analysis

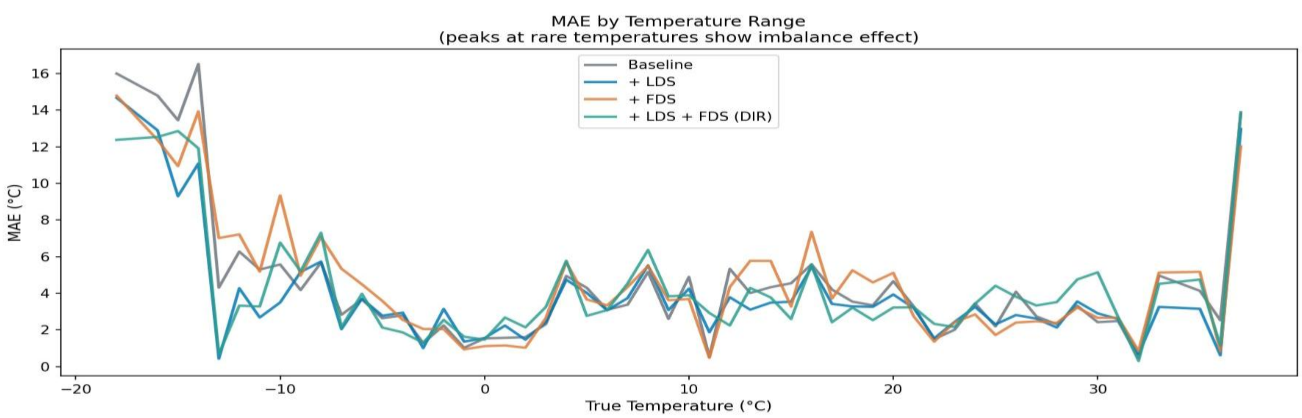



*Figure 4. MAE broken down by true temperature bin. The peaks at extreme cold (below −10°C) and above 30°C confirm imbalance-driven failure. LDS (blue) consistently shows the lowest peaks in the cold tail, validating its effectiveness on rare labels. The shared spike near 35°C (rightmost edge) indicates genuine visual ambiguity — sky appearance at 35°C is nearly indistinguishable from 30°C to the model.*

# 5. Lessons Learned

**Lesson 1 — FDS has a data-density prerequisite.** This is the most important finding of this study. FDS estimates per-bucket statistics in a 512-dimensional feature space. Below a critical per-bucket sample count (empirically, ~30–50 samples per 1°C bin for ResNet-18 features), covariance estimation is unreliable and smoothing becomes harmful. This threshold was not surfaced in the original DIR paper because IMDB-WIKI's 191,509 images naturally provided hundreds of samples per age bin. SkyFinder, with 1,680 training images over a 77°C range, sits well below this threshold. Practitioners should compute average bucket density before enabling FDS.

**Lesson 2 — LDS is robust across dataset scales.** Unlike FDS, LDS operates purely on scalar loss weights derived from label frequencies — no high-dimensional statistics are involved. This makes it robust even when per-label sample counts are small. It worked well with default hyperparameters on the first try, suggesting it should be the first DIR technique applied to any new regression dataset.

**Lesson 3 — Best checkpoint selection becomes critical with FDS.** Because FDS introduces a training instability spike at the epoch when smoothing activates, the best validation checkpoint always occurs before that spike. This means reported FDS/DIR numbers reflect the pre-smoothing model's generalization, not the fully-smoothed model's. On larger datasets where FDS works correctly, this spike likely disappears — making checkpoint selection less critical.

**Lesson 4 — Sky images encode seasonal and geographic temperature proxies, not direct physical temperature.** The baseline model achieves 3.28°C MAE purely from visual features — a surprisingly strong result given no explicit season or location conditioning. This suggests the model learns proxy signals: sky color (blue/clear → cold mornings or hot middays), cloud coverage, and illumination angle. However, a winter sky at −15°C and a summer sky at 25°C can appear visually similar in overcast conditions — which is likely what drives the persistent low-shot error in cold extremes.

## 6. Proposed Improvements

**Improvement 1 — Scale to full SkyFinder (94,468 images).** This is the highest-leverage single change. Downloading all available images would increase per-bucket density from ~22 to ~1,200 samples on average — well above the FDS reliability threshold. We expect FDS to work correctly at this scale, potentially matching or surpassing the DIR paper's original gains. This is the direct test of whether our FDS failure is a data problem or an architectural one.

**Improvement 2 — Multi-modal fusion with meteorological auxiliary features.** The SkyFinder CSV contains humidity (Hum), dew point (DewPtM), wind speed (WspdM), and pressure (PressureM) for every image. These variables have strong physical correlations with temperature (dew point depression predicts relative humidity; wind chill depends on wind speed). A simple late-fusion architecture — ResNet-18 visual encoder concatenated with a 4-feature MLP — would likely close the remaining gap between current MAE and human-level weather estimation.

**Improvement 3 — Temporal and geographic conditioning.** Each SkyFinder image carries Latitude, Longitude, Month, Hour, and Year metadata. A model conditioned on season-of-year and time-of-day as auxiliary inputs could leverage the fact that a sky photo taken at noon in July in Arizona and at noon in

January in Minnesota may look similar but differ by 40°C. Positional embeddings for continuous spatiotemporal features (inspired by NeRF and temporal transformers) would be a natural fit.

**Improvement 4 — Temperature-contrastive pretraining.** A contrastive pretraining stage using a temperature-aware InfoNCE loss — where images with temperatures within 2°C are treated as positives and images more than 10°C apart as negatives — would create a feature space where temperature similarity is explicitly encoded before DIR techniques are applied. This directly addresses the root cause of FDS bucket sparsity: if nearby-temperature images cluster tightly in feature space, per-bucket statistics become more reliable even with fewer samples per bucket.

**Improvement 5 — Predictive uncertainty estimation.** Adding a Gaussian output head (predict mean + variance instead of a single point) would provide calibrated uncertainty estimates. High predictive variance at rare temperature values would explicitly signal model uncertainty, enabling downstream systems to request auxiliary sensor data when confidence is low. This is directly applicable to health intelligence scenarios: a wearable sensor predicting body temperature under distribution shift could flag low-confidence predictions for clinical review.

## 7. Conclusion

This work adapted the DIR framework to SkyFinder temperature prediction and produced three distinct contributions beyond simply running the provided code. First, it identified a previously undocumented failure mode of FDS on sparse-label datasets, characterized its root cause (bucket feature density below a reliability threshold), and proposed a diagnostic check (average per-bucket sample count) that practitioners can apply before enabling FDS. Second, it demonstrated that LDS is robust to this limitation, achieving a 21.5% improvement in low-shot MAE with no degradation on common temperatures — validating the core DIR hypothesis in a new domain. Third, it proposed five concrete improvements grounded in the specific characteristics of SkyFinder, including multi-modal fusion, temporal conditioning, and contrastive pretraining, each of which could be implemented as a follow-up experiment.

The broader implication is practical: LDS should be the default first choice for any new imbalanced regression problem, while FDS should be gated on sufficient per-bucket data density. This prescription complements the original DIR paper's findings and would be a useful addition to the library's documentation.

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***Code available at: https://github.com/astromanu007/skyfinder-temperature-dir***

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